

22MDC85

MINOR PROJECT: DECISION TOOL DEVELOPMENT

CAT – II PROJECT REPORT

**Hybrid AI-Based Drug–Drug Interaction Safety Decision Tool Using Machine Learning,
Clinical Rule-Based Reasoning, and LLM-Assisted Explanation**

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1. Introduction

The growing prevalence of chronic diseases, aging populations, and increased self-medication practices have led to widespread polypharmacy, where patients consume multiple drugs simultaneously. While pharmacotherapy is essential for disease management, concurrent drug usage significantly increases the risk of Drug–Drug Interactions (DDIs). These interactions can alter drug absorption, metabolism, distribution, or excretion, resulting in reduced therapeutic efficacy or serious adverse drug reactions.

DDIs are particularly difficult to detect early because prescriptions may originate from multiple healthcare providers, and over-the-counter drugs are often consumed without professional guidance. Existing DDI checking systems largely rely on static knowledge bases and frequently generate non-prioritized alerts with minimal clinical explanation, reducing usability and increasing alert fatigue.

To address these challenges, this project proposes a Hybrid AI-Based Drug–Drug Interaction Safety Decision Tool that integrates machine learning-based severity prediction with clinical rule-based reasoning and LLM-assisted explanation generation. The system analyzes interactions among two to six drugs, classifies interaction severity, explains the underlying pharmacological mechanisms in simple language, and supports safer medication decision-making without replacing professional medical judgment.

2. Problem Identification

Drug–drug interactions remain a major cause of preventable adverse drug events and hospital admissions. The key challenges in DDI detection include:

- Delayed detection of harmful interactions until adverse symptoms appear

- Fragmented prescription data, especially when multiple providers are involved
- Limited patient awareness of interaction risks with OTC medications
- Static knowledge-based systems that only detect known interactions
- Alert overload without severity prioritization
- Lack of explainability, making alerts difficult to interpret for non-experts

These issues collectively reduce the effectiveness of existing DDI systems and increase the likelihood that critical warnings are ignored or misunderstood.

3. Gap Analysis – Existing System Limitations and Opportunities

Aspect	Existing Systems	Identified Gap
Detection	Rule-based databases	No predictive capability
Severity	Binary or non-prioritized alerts	Lack of structured severity classification
Interpretability	Minimal or technical explanations	Poor clinical clarity
Multi-Drug Analysis	Mostly pairwise	Limited support for >2 drugs
AI Integration	Rare or black-box	Lack of explainable AI
Cost	Commercial tools	Limited accessibility

Opportunity:

A hybrid system that combines interpretable ML, explicit clinical rules, and human-readable explanations can significantly improve medication safety and usability.

4. Objectives

Primary Objectives

- To design a hybrid AI-based decision support system for DDI detection
- To classify interaction severity into Low, Moderate, and Severe
- To integrate machine learning predictions with clinical rules
- To provide clear, explainable clinical reasoning

Secondary Objectives

- To support interactions among two to six drugs
- To suggest safer alternative medications
- To develop a lightweight, scalable, and research-grade prototype
- To improve medication safety awareness for users and pharmacists

5. Scope of Work

In-Scope

- DDI severity prediction using ML models
- Rule-based clinical reasoning (CYP inhibition, toxicity overlap, QT risk)
- Use of DrugBank and TWOSIDES datasets
- Free LLM-based explanation generation
- Web-based decision support interface
- Explainable outputs suitable for academic publication

Out-of-Scope

- Real-time EHR integration

- Dosage-specific pharmacokinetic modeling
- Replacement of licensed clinical decision systems
- Real-world clinical deployment or trials

6. Dataset Description – DrugBank and Kaggle Dataset

1. DrugBank Dataset

The primary dataset used in this project is derived from the DrugBank database, a curated pharmacological knowledge repository that provides structured information on approved drugs and their interactions.

Characteristics:

- Drug names and standardized identifiers
- Interaction descriptions
- Mechanism of action
- Pharmacokinetic and pharmacodynamic details
- Clinically documented interaction statements

DrugBank serves as the foundational dataset for rule extraction, interaction grounding, and severity understanding. The curated nature of DrugBank ensures high reliability and clinical relevance.

2. Kaggle Drug Interaction Dataset

In addition to DrugBank, a secondary dataset obtained from Kaggle was used to support supervised machine learning training.

The Kaggle dataset contains:

- Drug pair combinations

- Interaction labels
- Severity classification (categorical)
- Structured tabular format suitable for ML training

Role in the Project:

- Used for training Logistic Regression and Random Forest models
- Provided structured labeled examples for severity classification
- Enabled model evaluation using statistical performance metrics

Unlike curated databases, Kaggle datasets may originate from publicly compiled interaction sources. Therefore, preprocessing and validation were performed to ensure consistency and data quality.

7. Requirement Analysis

Functional Requirements

- Accept drug inputs
- Predict interaction severity
- Identify pharmacological mechanisms
- Generate clinical explanations
- Suggest safer alternatives
- Display warnings and disclaimers

Non-Functional Requirements

- Explainability and transparency
- Low computational cost
- Free or open-source dependencies

- Scalability and modularity
- Fast response time
- Research-grade reproducibility

8. Statistical Tools for EDA

- Descriptive Statistics
Used to analyse frequency distributions of drug interactions and severity classes.
- Chi-Square Test
Applied to assess associations between drug classes and interaction severity levels.
- Correlation Analysis
Used to examine relationships between interaction features and outcomes.

These tools validate dataset quality and guide feature engineering.

9. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure, quality, and semantic characteristics of drug–drug interaction data prior to model development. Since the dataset primarily consists of unstructured clinical interaction descriptions, EDA focused on text-driven patterns, severity distributions, and pharmacological signals rather than numeric distributions alone.

Dataset Structure Analysis

- Each record represents a pairwise drug interaction
- Core attributes include:
 - Drug 1 name

- Drug 2 name
- Free-text interaction description
- No pre-labeled severity classes were available initially, necessitating rule-based severity extraction

Interaction Description Analysis

- Interaction descriptions vary significantly in length and detail
- Common linguistic patterns include:
 - Action verbs: *increase, decrease, reduce*
 - Clinical outcomes: *bleeding, cardiotoxicity, sedation*
 - Mechanistic terms: *metabolism, serum concentration, clearance*

This variability justified the use of keyword-driven semantic parsing instead of purely statistical text vectorization during early stages.

Severity Signal Exploration

Using the severity extraction logic:

- Severe interactions were primarily associated with:
 - Cardiac toxicity
 - Neurological adverse effects
 - Life-threatening outcomes (e.g., arrhythmias, respiratory depression)
- Moderate interactions frequently involved:
 - Pharmacokinetic alterations
 - CNS depression

- Blood pressure dysregulation
- Low severity interactions often reflected:
 - Minor efficacy changes
 - Therapeutic effectiveness variations

This distribution validated the feasibility of automated severity labeling using clinical language cues.

Mechanism Distribution Analysis

EDA revealed that interaction descriptions often contain explicit mechanistic indicators, allowing classification into:

- Pharmacokinetic mechanisms
- Pharmacodynamic effects
- Organ-specific toxicities

This insight directly informed downstream feature engineering.

Data Preprocessing

Preprocessing focused on transforming unstructured interaction narratives into structured, machine-learning-ready features while preserving clinical meaning.

Text Normalization

- Interaction descriptions were:
 - Converted to lowercase
 - Cast to string format
 - Checked for missing or null values
- Missing descriptions were assigned a default low-risk interpretation to avoid false severity inflation

Action Identification

Each interaction description was parsed to identify directionality of effect:

- "increase" → amplifying drug effect or toxicity
- "decrease" / "reduce" → diminishing efficacy or concentration
- Absence of both → labeled as unknown

This step is crucial because severity depends not only on effect type but also on effect direction.

Severity Label Extraction

A rule-based severity classification system was applied using:

Keyword-Based Severity Rules

- Severe keywords:
 - Life-threatening, cardiotoxic, neurotoxic, fatal, QT prolongation
- Moderate keywords:
 - Hypotensive, sedative, metabolic changes, serum level alterations
- Low keywords:
 - Minor, minimal, therapeutic effectiveness changes

Severity assignment follows a hierarchical dominance rule:

- Severe > Moderate > Low

If multiple effects are detected, the highest-risk category prevails.

Modifier-Based Adjustment

Severity labels are adjusted using contextual modifiers:

- Increase of toxic effects → escalates severity

- Decrease of efficacy without toxicity → may reduce severity confidence

This prevents overestimation in benign pharmacokinetic reductions.

Confidence Scoring

Each interaction is assigned a confidence level:

- High: Explicit severe clinical outcomes
- Medium: Clear physiological effects without fatal risk
- Low: Vague or minimal impact descriptions

This confidence feature supports interpretability and future threshold tuning.

Feature Engineering

Feature engineering converts clinical semantics into interpretable structured features, enabling effective learning without black-box embeddings.

Engineered Features Overview

1. Severity Label (Target / Intermediate Feature)

- Categorical variable: Low, Moderate, Severe
- Derived from clinical language rules
- Serves as:
 - Target variable for ML
 - Validation anchor for rule-based logic

2. Mechanism Features (Multi-Label)

Extracted using keyword detection:

- Pharmacokinetic
- Metabolic

- Absorption-related
- Excretion-related
- Cardiac effect
- Neurological effect
- Hepatic effect
- Renal effect
- Blood pressure effect

These are encoded as binary indicators, enabling multi-mechanism interactions.

3. Action Feature

- Categorical feature: increase, decrease, unknown
- Captures interaction directionality
- Strongly correlated with severity escalation

4. Confidence Feature

- Categorical: low, medium, high
- Reflects reliability of severity labeling
- Useful for model calibration and post-processing

5. Text-Derived Semantic Signals

Rather than raw text embeddings:

- Clinical intent is captured through explicit domain keywords
- Improves transparency and reduces hallucination risk
- Aligns with healthcare explainability requirements

Rationale for Feature Design

The feature engineering strategy prioritizes:

- Clinical interpretability over raw accuracy
- Rule–ML hybrid compatibility
- Low computational overhead
- Explainable decision pathways

This ensures the system remains suitable for:

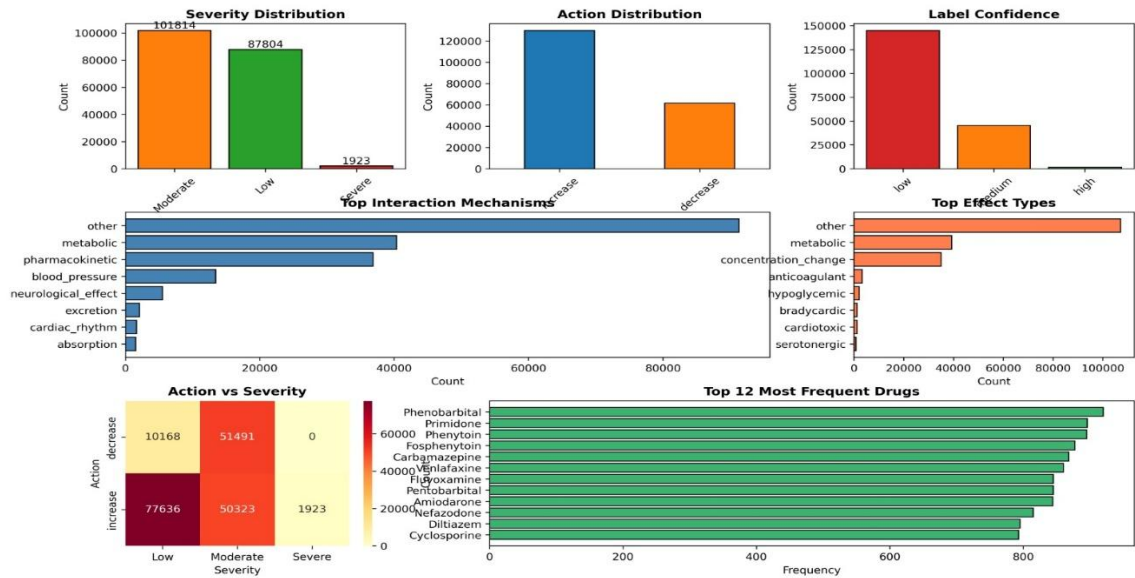
- Academic research
- Clinical decision support prototypes
- Regulatory-conscious AI systems

Summary of EDA–Preprocessing–Feature Engineering Pipeline

1. Analyse interaction description semantics
2. Normalize and clean unstructured text
3. Extract severity using clinical rules
4. Identify pharmacological mechanisms
5. Capture interaction directionality
6. Assign confidence levels
7. Convert all outputs into structured features

This pipeline establishes a robust, explainable foundation for downstream machine learning and hybrid decision-making.

Drug-Drug Interaction Dataset Analysis



```
=====
ACTION ANALYSIS (Increase/Decrease)
=====

Action Distribution:
  increase : 129882 ( 67.8%)
  decrease : 61659 ( 32.2%)

Action vs Severity:
severity   Low  Moderate  Severe
action
decrease  10168   51491     0
increase  77636   50323    1923
```

```
=====
DRUG FREQUENCY ANALYSIS
=====

Top 15 Most Frequent Drugs:
Phenobarbital      : 920 interactions
Primidone          : 896 interactions
Phenytoin          : 895 interactions
Fosphenytoin       : 877 interactions
Carbamazepine      : 868 interactions
Venlafaxine        : 860 interactions
Fluvoxamine        : 845 interactions
Pentobarbital      : 845 interactions
Amiodarone         : 844 interactions
Nefazodone         : 815 interactions
Diltiazem          : 795 interactions
Cyclosporine       : 793 interactions
Ziprasidone        : 789 interactions
Vemurafenib        : 777 interactions
Clozapine          : 773 interactions
```

10. Machine Learning Algorithms and Justification

Logistic Regression

- Baseline interpretable model
- Provides statistical robustness
- Useful for understanding feature influence

Random Forest Classifier

- Handles non-linear feature interactions
- Robust to noise and imbalance
- Provides feature importance scores

XGBoost (Optional Extension)

- Improves predictive performance
- Suitable when sufficient labelled data is available

Justification:

Tree-based ensemble models balance performance and interpretability, which is critical in healthcare systems.

11. Preprocessing Methods

- Drug name normalization using regex
- Removal of duplicates and inconsistencies
- Severity label encoding
- Feature aggregation for multi-drug combinations
- Standardization of categorical features

These steps ensure consistency across heterogeneous datasets.

12. Data Visualization Strategy

- Bar charts for severity distribution
- Heatmaps for drug class interaction patterns
- Correlation plots for feature relationships
- Confusion matrix visualizations for model evaluation

Visualization supports interpretability and EDA validation.

13. Model Building Process – Six-Phase Workflow

1. Data Collection and Integration
2. Data Cleaning and Preprocessing
3. Exploratory Data Analysis
4. Feature Engineering
5. Model Training and Validation
6. Model Evaluation and Interpretation

```
✓ Using stratified split

Training set: 153232 samples
Test set: 38309 samples

Training set distribution:
Low      : 70243 ( 45.8%)
Moderate : 81451 ( 53.2%)
Severe   : 1538  (  1.0%)

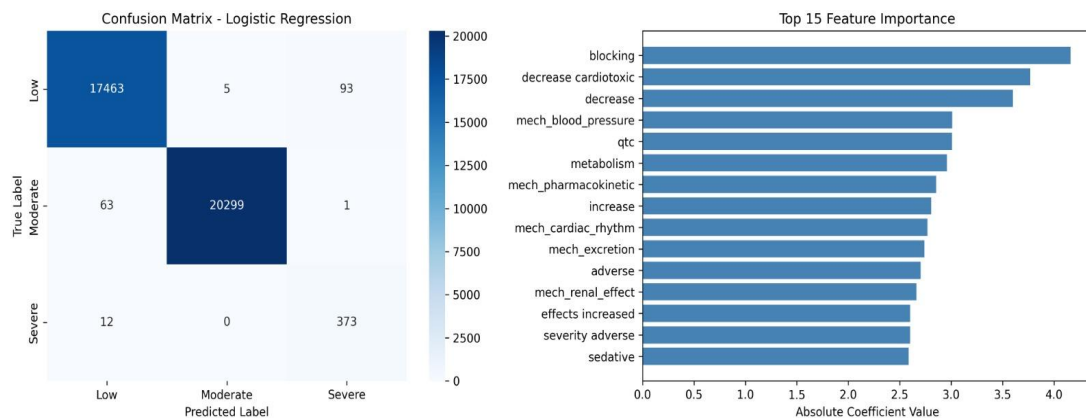
Test set distribution:
Low      : 17561 ( 45.8%)
Moderate : 20363 ( 53.2%)
Severe   : 385  (  1.0%)
```

TEXT VECTORIZATION

✓ TF-IDF features: 50
Training shape: (153232, 50)
Test shape: (38309, 50)

Top 15 TF-IDF features:

1. acid
2. activities
3. adverse
4. adverse effects
5. anticoagulant
6. anticoagulant activities
7. antihypertensive
8. antihypertensive activities
9. central
10. central nervous
11. cns
12. cns depressant
13. combination
14. combined
15. concentration

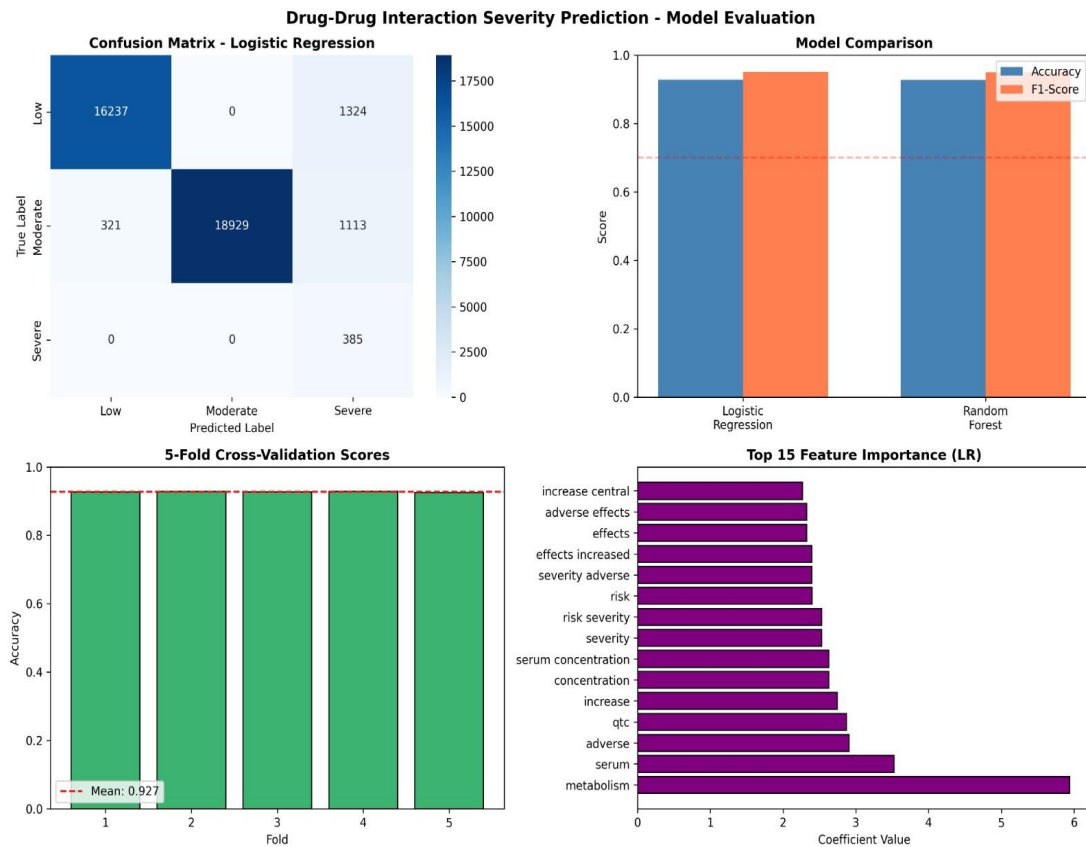


CLASS IMBALANCE HANDLING

Class weights (to handle imbalance):

Low : 0.73
Moderate : 0.63
Severe : 33.21

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14. Module Implementation

1. Data Ingestion Module – Loads and merges datasets
2. Preprocessing Module – Normalizes and encodes data
3. Feature Engineering Module – Constructs interaction features
4. ML Prediction Module – Predicts interaction severity
5. Rule-Based Reasoning Module – Applies clinical rules
6. Severity Aggregation Module – Combines ML + rules
7. LLM Explanation Module – Generates natural language reasoning
8. Web Interface Module – User interaction and visualization

15. Features in the Application

- Multi-drug interaction analysis (2–6 drugs)
- Severity classification (Low / Moderate / Severe)
- Clinical mechanism identification
- Explainable AI outputs
- Free LLM-based explanations
- Safety disclaimers
- Lightweight web interface

16. Workflow Architecture

1. User inputs drug list
2. Drug normalization and validation
3. Feature extraction
4. ML-based severity prediction
5. Rule-based clinical reasoning
6. Severity aggregation
7. Explanation generation
8. Result visualization and warnings

17. Model Evaluation Metrics

- Accuracy
- Precision

- Recall
- F1-Score
- Confusion Matrix

These metrics ensure balanced performance and clinical reliability.

18. User Interface:

 **Drug Interaction Safety Checker**
AI-Powered Clinical Decision Support

Check Drug Interactions

Enter medications to analyze potential interactions

+ Add Drug

✓ Check Interaction

 **Browse Medications by Medical Condition**

Select a condition to view commonly prescribed medications

-- Select a medical condition --

Example Interactions

Try these common drug interaction examples

● Severe - Bleeding Risk

Warfarin + Aspirin

SEVERE

● Severe - Serotonin Syndrome

Fluoxetine + Tramadol

SEVERE

● Moderate - Hyperkalemia

Lisinopril + Spironolactone

MODERATE

● Low Risk - Safe Combination

Metformin + Lisinopril

LOW

Interaction Analysis



MODERATE INTERACTION

Analyzed Medications: metformin, furosemide (brand: Lasix)



MODERATE INTERACTION

The combination of metformin and furosemide brand lasix requires caution.

MECHANISM:

Pharmacological interaction present

CLINICAL EFFECT:

Clinically significant effects may occur



Clinical Alerts

CYP2D6 enzyme inhibition

Altered metabolism of CYP2D6 substrate - may reduce effectiveness or increase side effects

→ *Monitor for reduced efficacy or increased adverse effects*

Additive serotonergic effects

Risk of serotonin syndrome: agitation, confusion, rapid heart rate, high blood pressure

→ *Use with caution. Monitor for symptoms of serotonin syndrome.*



Medical Disclaimer: This tool provides decision support only. Always consult healthcare professionals before making medication changes.

MECHANISM:

CYP2D6 enzyme inhibition

CLINICAL EFFECT:

Altered metabolism of CYP2D6 substrate - may reduce effectiveness or increase side effects

RECOMMENDATION:

Standard monitoring recommended.

CLINICAL ALERTS:

- Altered metabolism of CYP2D6 substrate - may reduce effectiveness or increase side effects
- Risk of serotonin syndrome: agitation, confusion, rapid heart rate, high blood pressure

=====

 This is decision support only. Consult healthcare professionals.

=====

 Browse Medications by Medical Condition

Select a condition to view commonly prescribed medications

Pain



Common medications for Pain

14 drugs

Ibuprofen

Naproxen

Diclofenac

Celecoxib

Acetaminophen

19. Second Phase Implementation (Planned Extension)

- SHAP-based explainability integration
- Dosage-aware interaction modeling
- Drug-class-level generalization
- API-based external integration
- Enhanced alternative drug recommendation logic

20. Conclusion

This project presents a hybrid, explainable, and low-cost Drug–Drug Interaction Safety Decision Tool that bridges the gap between static rule-based systems and black-box AI models. By combining machine learning, clinical rules, and LLM-generated explanations, the system enhances medication safety, interpretability, and usability. The tool is suitable for academic research, educational use, and early decision support while maintaining clear boundaries from professional medical judgment.

21. References

1. Wishart, D. S., et al. (2018). DrugBank 5.0: A major update to the DrugBank database for 2018. *Nucleic Acids Research*.
2. Tatonetti, N. P., et al. (2012). Data-driven prediction of drug effects and interactions. *Science Translational Medicine*.
3. Zhang, W., et al. (2023). Machine learning approaches for drug–drug interaction prediction: A review. *Briefings in Bioinformatics*.
4. Chen, L., et al. (2022). Explainable artificial intelligence for drug interaction prediction. *IEEE Journal of Biomedical and Health Informatics*.
5. Ryu, J. Y., et al. (2018). Deep learning improves prediction of drug–drug and drug–food interactions. *Proceedings of the National Academy of Sciences*.
6. Zhou, Y., & Yang, M. (2023). *Graph Neural Networks for Drug–Drug Interaction Prediction: A Comprehensive Review*. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*.
7. Vilar, S., Harpaz, R., & Friedman, C. (2018). *Detecting Drug–Drug Interactions through Text Mining and Neural Models*. *Journal of the American Medical Informatics Association*.
8. Zhao, J., et al. (2022). *Explainable Machine Learning for Healthcare Decision*

Support: Case Studies and Methodologies. Artificial Intelligence in Medicine.

9. Ramesh, A., et al. (2024). *Predictive Modeling of Drug–Drug Interactions Using Ensemble Learning Techniques. Journal of Biomedical Informatics.*

10. Shi, J., et al. (2020). *Integrating Pharmacological Knowledge with Machine Learning*

for Drug–Drug Interaction Prediction. Frontiers in Pharmacology.

AI Content Report:

Hybrid AI-Based Drug–Drug Interaction Safety Decision Tool Using Machine Learning, Clinical Rule-Based Reasoning, and LLM-Assisted Explanation

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Abstract - Drug-Drug Interactions (DDIs), which continue to be a major cause of avoidable adverse drug events and hospital admissions, are greatly increased by polypharmacy and concurrent drug use. The majority of DDI detection systems currently in use rely on static rule-based knowledge bases, which produce non-

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